Costly Information and Beliefs: A Direct Test of Rational Inattention Models

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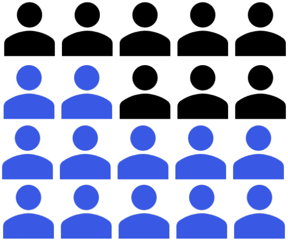
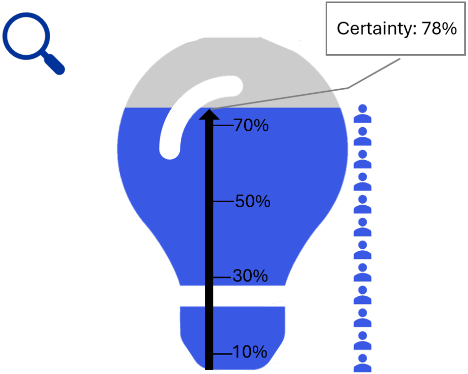
What will you do? – Detective's Dilemma

- Suspicious report received
- Uncertain: "Culprit is in town" "No culprit in town"

Motivation
Motivating Example





How many people should the detective interview?

vs.

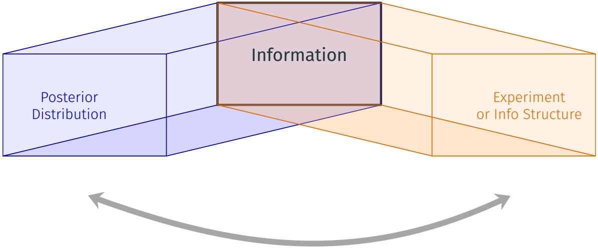
How certain should the detective be before concluding there is no culprit?

Motivation
Theories of an Inattentive DM

Rational Inattention Theory:

$$\max_{\text{Info} \in I} \mathbb{E}[\text{Revenue}|\text{Info}] - \text{Cost}(\text{Info})$$

✔ Different Representations: *Info*



- Mathematically equivalent – Bayes plausibility
  - From the analyst's perspective

Rational Inattention Theory:

$$\max_{\text{Info} \in I} \mathbb{E}[\text{Revenue}|\text{Info}] - \text{Cost}(\text{Info})$$

✔Traditional Representation: *Info = Posterior Distributions*
$\hookrightarrow$ (Distribution of Posteriors over State Space $\in \Delta(\Delta(\Omega))$)

Why?
Recent theoretical models
– To infer the structure of information costs from observed behavior, assuming the "Consistency" b/t representations

- Hébert and Woodford (2021); Caplin et al. (2022)
- Pomatto et al. (2023); Bloedel and Zhong (2025)

Experiment-based and posterior-based costs are **not generally interchangeable!** (Denti et al. (2022))
Mere Framing Effect? – No?!

Motivation
Inattention Experiments tasks

In the Lab:

Guess correct state $\Rightarrow$ win prize

✔Observable variable:
Choice given State (State Dependent Stochastic Choice)
$\hookrightarrow$ Approach: (Caplin and Dean (2015)) Reconstruct optimal posterior distr. from SDSC

- Test and estimate models of inattention using SDSC
- Caplin et al. (2019); Dewan and Neligh (2020); Dean and Neligh (2023)

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$\Rightarrow$ *"What is the appropriate decision variable to represent inattention?"*

Possible decision variables from Theory:
“Optimal Posterior Distribution” or “Optimal Information Structure”

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“Optimal Posterior Distribution” or “Optimal Information Structure”

$\downarrow$

Implemented as choosing:
“Desired Accuracy” or “Desired Sample size”

Main research questions

Assuming DM is optimizing

$$\max_{\text{Info}\in I} \mathbb{E}[\text{Revenue}|\text{Info}] - \text{Cost}(\text{Info}),$$

1. Are the two frames "practically" *equivalent*?

$\rightarrow$ Choice of "Posterior distr. (Accuracy)"
$\approx$ Choice of "Info Structure (Sampling amount)"

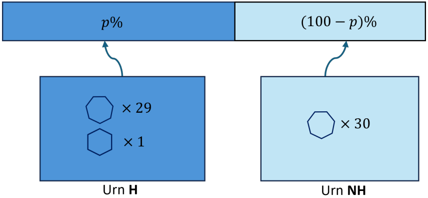
2. *How* is information costly?

- Based on "Posterior distr. (Accuracy)"
(Posterior-based treatment)
- Based on "Info Structure (Sampling amount)"
(Experiment-based treatment)

3. An individual's consistent process of info representation?

$\rightarrow$ Let them choose again between the previous choices

Choice of "Posterior distr. (Accuracy)"
vs. Choice of "Info Structure (Sampling amount)"



Decision Problem

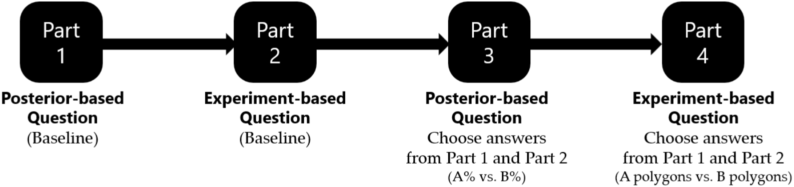
- Task = Guess the Urn
- Two Urns: $\Omega = \{H, NH\}$
  - $p \equiv Pr(H)$: a prior belief
- Two Action: $A = \{H, NH\}$
  - $u(NH, NH) = u(H, H)$
    $= r \times u(\$20) > 0$
  - $u(NH, H) = u(H, NH) = 0$
- "Inspect the urn" before making guess
  - Draw polygons sequentially from the selected urn *without replacement*
  - Each polygon is shown for 0.7 sec

Design
Treatments

Experiment 1

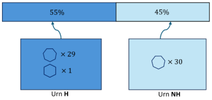
Treatment 1	Treatment 2	Treatment 3
Posterior-based	Experiment-based	Experiment-based+

Experiment 2

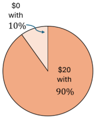


Round 1 of 168

The chance of URN **H** being selected is **55%**.
The chance of URN **NH** being selected is **45%**.



Correct guess gives you **90%** chance of winning $20 .



How likely do you want the guess NH to be correct in case no hexagon appears?

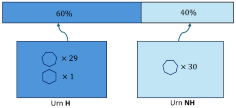
Next

How likely do you want the guess NH to be correct in case no hexagon appears?

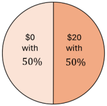
Design
Experiment-based Treatment

Round 1 of 168

The chance of URN **H** being selected is **60%**.
The chance of URN **NH** being selected is **40%**.



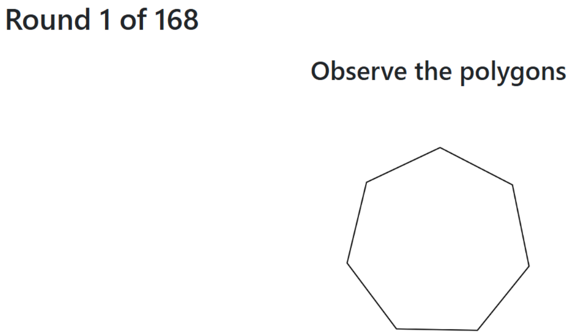
Correct guess gives you **50%** chance of winning $20 .



How many polygons would you like to take out of the URN?

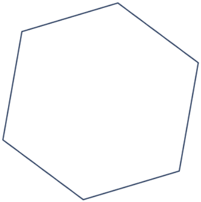
Next

How many polygons would you like to take out of the URN?



Round 2 of 168

Observe the polygons



Design
Guessing phase

Guess the URN

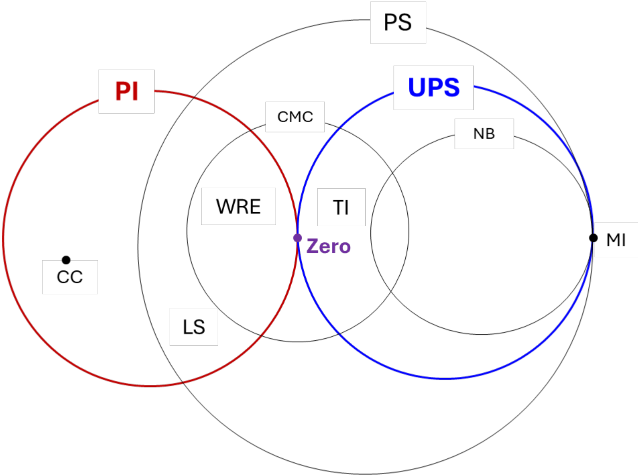
You guess that the selected Urn for this round is

○ No Hexagon ○ Hexagon

Next

- Simple task
  - "*Black Swan*" Environment:
    A **Posterior Belief** uniquely indices a Posterior Distribution
    A **Sampling fraction** uniquely indices an Info Structure
  - Operationalize "posterior distributions" and "information structures"
- Emphasizes paying attention
- Directly observe desired informativeness
- Identify behavioral models
  - Vary prior p, reward r

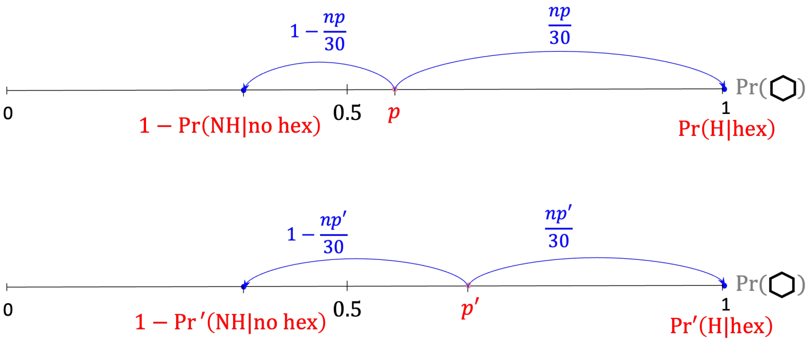
Strzalecki (2025)'s diagram



UPS ∩ PI = Zero Cost
(Denti et al. (2022); Bloedel and Zhong (2025))

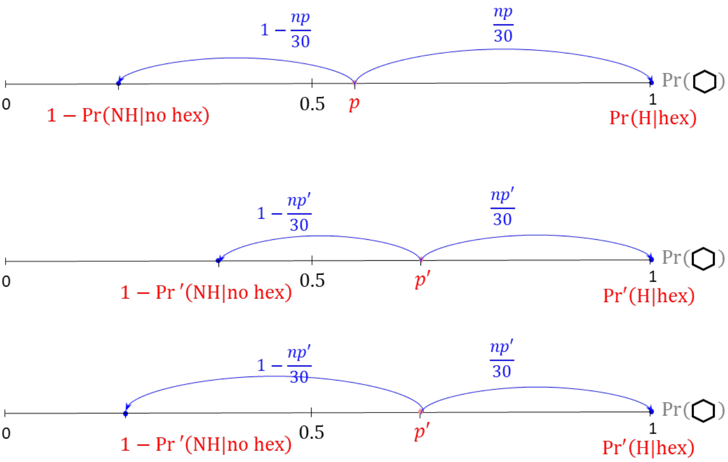
Analysis Background

Classifying Behaviors: Uniform Posterior Separable Class



- Locally Invariant Posteriors (Caplin et al., 2022)
- Following Dean and Neligh (2023)

Analysis Background
Classifying Behaviors: Prior Invariant Class

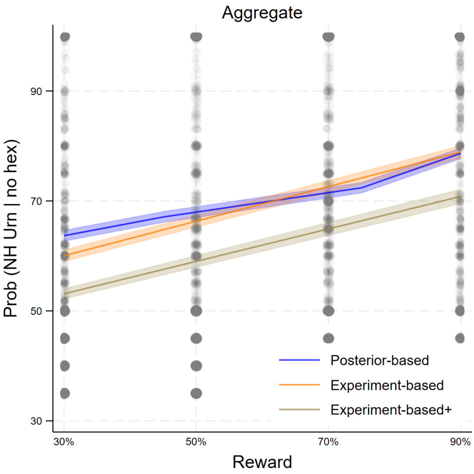


$$\underbrace{r\left(\frac{np}{30}\times 1+\left(1-\frac{np}{30}\right)\Pr(NH|\text{no hex})\right)}_{\text{Revenue}}-\underbrace{C(\Pr(NH|\text{no hex}))}_{\text{Cost}}$$

Procedures

- Ohio State University students, recruited via ORSEE, Lab experiment
- Posterior-based: 50 subjects; Experiment-based: 41; Experiment-based+: 34;
- Payments: Randomly picked two rounds for their guess (20 USD)
- Average of 27.94 USD (including 10 USD show-up)
- Range: 30-120 mins
- Total observations: 21,000

Results
Rationally Inattentive?



Aggregate: Accuracy increases as reward increases

Results
Rationally Inattentive? (Individuals)

$$\text{Prob(NH Urn}|\text{nh}) = \beta_0 + \beta_1 \text{reward} + \beta_2 \text{prior} + \epsilon$$

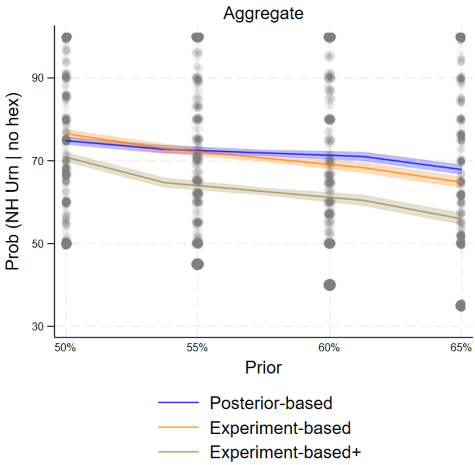
	$\beta_1 \geq 0$	$\beta_1 > 0$	$\beta_1 < 0$
Posterior-based	50	24	0
Experiment-based	40	23	1
Experiment-based+	34	19	0
All Sessions	124	66	1

Table 1: Responsiveness to rewards

Result 1:

Almost all subjects are rationally inattentive.
About half are strictly responsive to the reward.

Results
How is Info Costly? (Individuals)



$$\text{Prob(NH|nh)} = \beta_0 + \beta_1 \text{reward} + \color{red}{\beta_2}\text{prior} + \epsilon$$

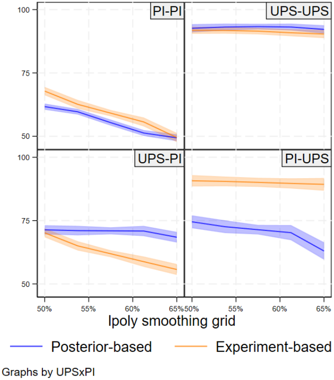
	$\beta_2 < 0$	$\beta_2 = 0$			
	PI	UPS	Zero	High	Neither
Post-based	20	25	2	2	1
Exp-based	26	8	5	1	1 (not RI)
Exp-based+	25	7	0	2	0
Total	71	40	7	5	2

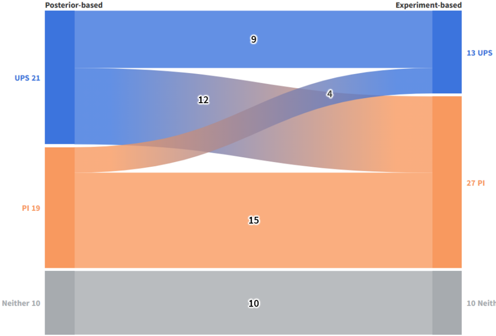
Table 2: Inattention Models

Result 2:

Both UPS and PI are empirically valid, as well as Zero Cost model.
The dominant behavior changes depending on the representations.

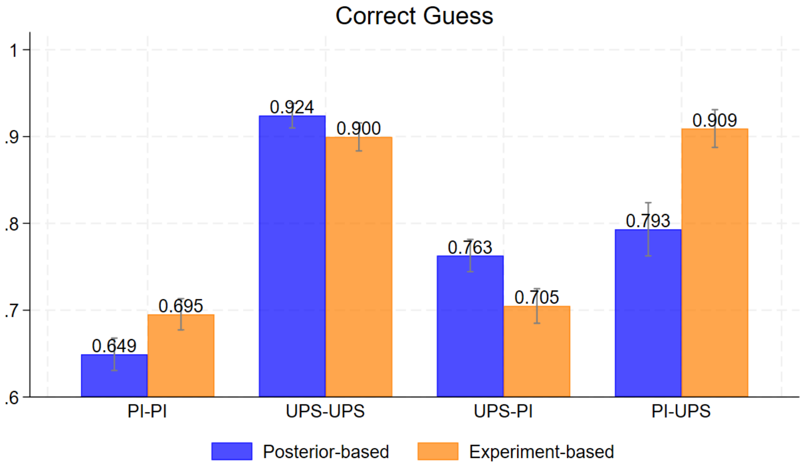
Results
Is Behavior Consistent?





- Heterogeneity in response to framing change
- Switchers: UPS to PI

Results
Consistent Individuals



- UPS more accurate than PI
- PI more accurate in Exp-based, UPS in Post-based

Result 3

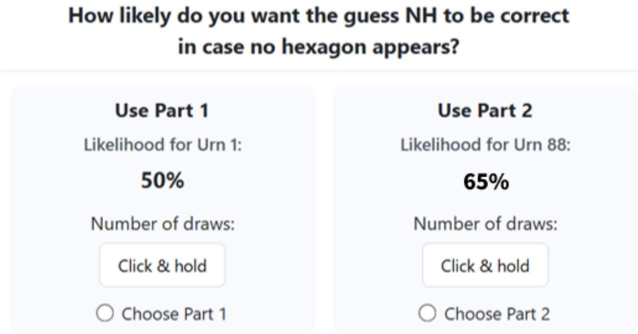
Majority of subjects (24 of 40) are consistent.
Among switchers, a significant portion switch (12 of 16) mostly from UPS to PI.
Framing affects accuracy regardless of consistency.

+ This is NOT due to the failure in Bayesian updating!

Belief mapping does not improve consistency.
Framing affects behaviors, not failure in Bayesianism.

Exp-based+ Frame Treatment

Results
Persistent framing in mind?



			Noisy (Random)
Post-based	Persistent	24	7
	Frame-dependent	6	
Exp-based	Persistent	18	
	Frame-dependent	2	

Result 5

"Correct" framing for each individual?
→Largely dependent on the representation

Conclusion

1. How is information costly?

   - Both UPS and PI models are **empirically valid** in terms of both **direct** (Posterior) and **natural** (Info Structure) frames.

2. Are the decision problems under the two framings "practically" equivalent? **NO**

   $\rightarrow$ Can affect accuracy

   $\rightarrow$ Can induce to switch behavior

   $\rightarrow$ Belief mapping DOES NOT improve consistency.

3. What is the "correct" framing for each individual?

   $\rightarrow$ Largely dependent on the representation

   - How should we incorporate framing dependence into the models and rationalize such behaviors?

Thank you for your attention!

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P.S. I will be on the 2026-2027 Job Market

References

Bloedel, A. W. and Zhong, W. (2025). The Cost of Optimally Acquired Information. arXiv:2511.05466 [econ].

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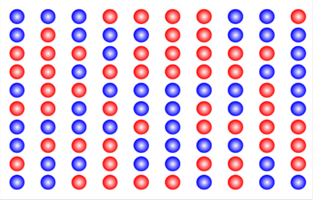
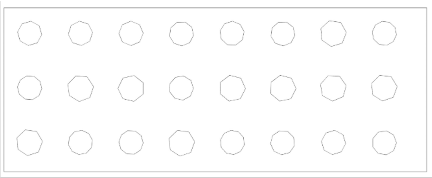
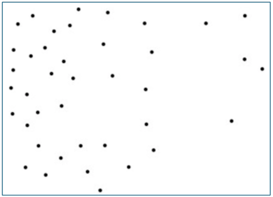


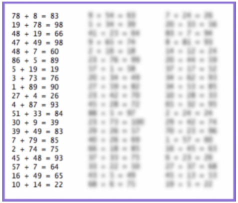
Dean & Neligh (2023, JPE)



Caplin et al. (2020, QJE)



Dewan & Neligh (2020, JET)



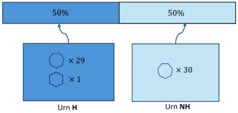
Ambuehl et al. (2025, RESTAT)

Figure 1: Attention Tasks

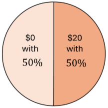
Robustness?
Consistency: Belief mapping?

Round 1 of 168

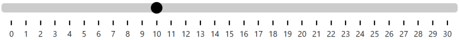
The chance of URN H being selected is 50%.
The chance of URN NH being selected is 50%.



Correct guess gives you 50% chance of winning $20 .



Please click the bar below to reveal the slider to indicate the polygons to be taken out.



Taking out **10** polygons
The likelihood of the Guess NH to be correct in case no hexagon appears: **60.00%**

How many polygons would you like to take out of the URN?

Next

Results

Exp-based+ likelihood

	Exp-based+ UPS	Exp-based+ PI
Post-based UPS	5	12
Post-based PI	2	14
Consistent	19	
Switcher	14	

Framing affects behaviors, not failure in Bayesianism

Result 4

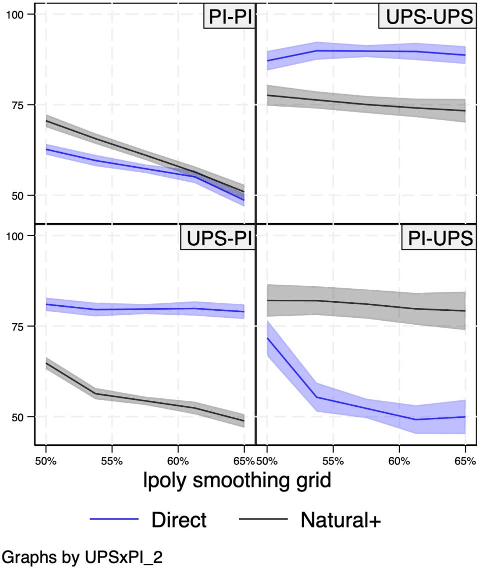
Belief mapping does not improve consistency.
Effect may be purely from framing.

1. Binary state, Binary action case: $\Omega = \{0, 1\}, A = \{0, 1\}$.
    - DM willing to match the action to the state:
      $r \equiv u(0,0) = u(1,1) > u(a,\omega)$, with $a \neq \omega$ where $a, \omega \in \{0,1\}$.
2. Information technology
    - Two types of signals, γ_0 and γ_1,
    - Signal γ_1, fully reveals the state:
      $\gamma_1(w=0) \equiv \mathsf{Pr}(\omega = 0|\gamma_1) = 0, \gamma_1(w=1) \equiv \mathsf{Pr}(\omega = 1|\gamma_1) = 1$.

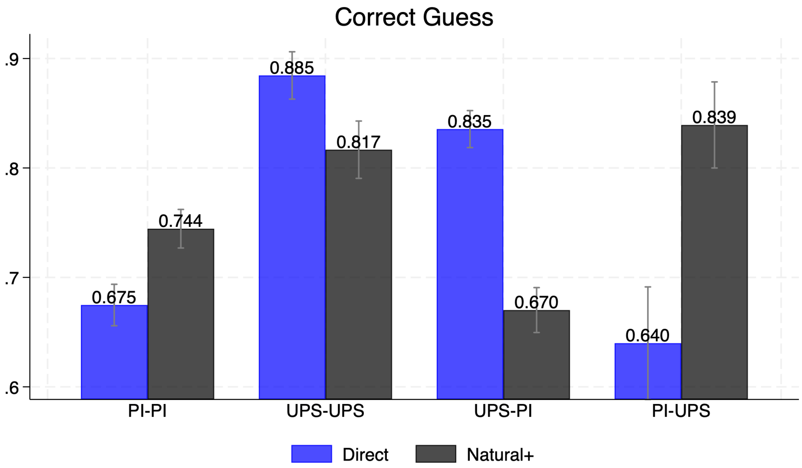
$$\pi(x) = \begin{array}{c c} & \begin{array}{cc} \gamma=\gamma_0 & \gamma=\gamma_1 \end{array} \\ \begin{array}{c} \omega=0 \\ \omega=1 \end{array} & \begin{pmatrix} 1 & 0 \\ 1-x & x \end{pmatrix} \end{array}$$

3. DM chooses the informativeness that is enough to choose the action 0 after seeing signal γ_0,
    - Only allowing: $x = 0$,
      x s.t. $\gamma_0(0) \geq \frac{1}{2}$, or equivalently $x(p) \geq 1 - \frac{1-p}{p}$.
    - Action-relevant information

Is Behavior Consistent w/ mapping?



Is Behavior Consistent w/ mapping?



Result 4:

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